



Lecture 2 – Part 1: NSL-KDD Dataset

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Agenda

- Announcements
- Course Dataset: NSL-KDD

Announcements

- HW-1 is released after today's lecture (5pm ET).
 - Read its instructions carefully.
 - Pay attention to the unique identifications required to be included in every screenshot.
 - Read the permitted AI usage section carefully.
 - Refer to the published tutorials under the "Modules" section on Canvas to help you with Part-1.
 - HW-1 is due in two weeks.
- System setup recitation office hours are held this Friday. Refer to last lecture slides for the timing and logistics.

NSL-KDD Dataset

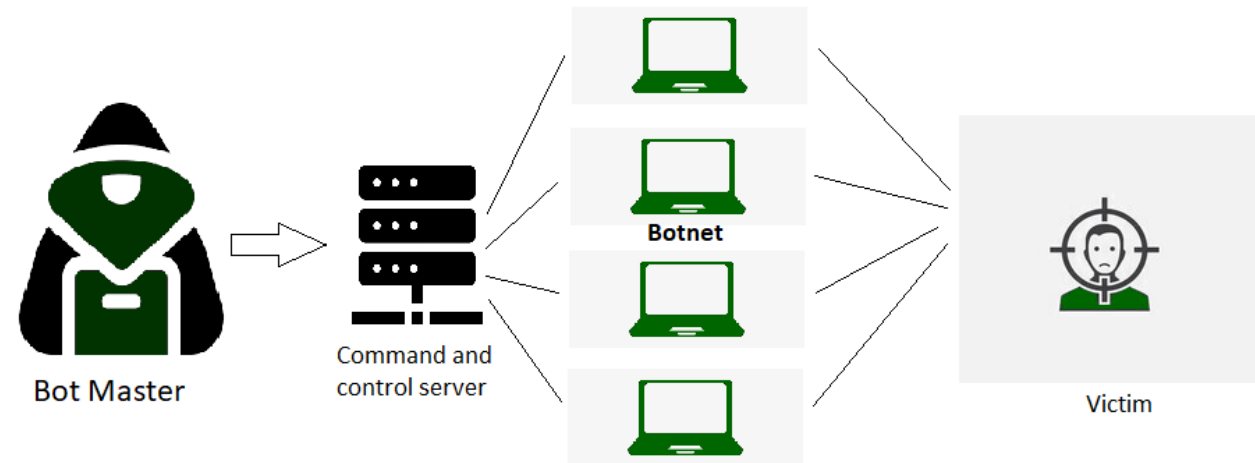
- The NSL-KDD dataset contains 43 features per record, with 41 of the features referring to the traffic input itself and the last two columns represent the **activity type** (whether it is normal or attack) and **Score/Difficulty level** (the number of classifiers that successfully classified that record out of 21 total).



- Let's start by exploring the output. The **score/difficulty level column** takes an integer value up to 21. On the other hand, the **activity type column** indicates either normal or type of the attack.
- In the dataset, there are 4 different classes of attacks:
 - Denial of Service (DoS)
 - Probe
 - User to Root (U2R)
 - Remote to Local (R2L)

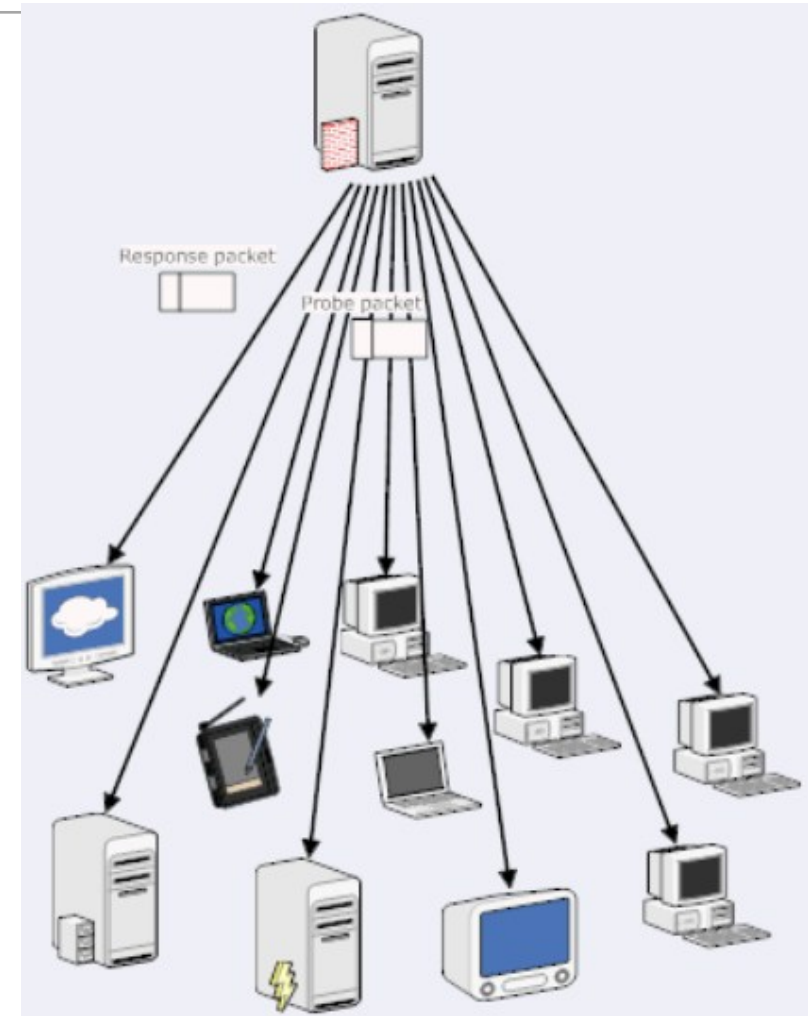
NSL-KDD Dataset – DoS Attacks

- DoS is an attack that tries to shut down the traffic flow to and from the target system. The Intrusion Detection System (IDS) is flooded with an abnormal amount of traffic, which the system can't handle, and shuts down to protect itself. This prevents normal traffic from visiting a network. An example of this could be an online retailer getting flooded with online orders on a day with a big sale, and because the network can't handle all the requests, it will shutdown and therefore, prevents paying customers from purchasing anything. This is the most common attack in the data set.



NSL-KDD Dataset – Probe Attacks

- Probe or surveillance is an attack that tries to get information from a network.
- The goal here is to act like a thief and steal important information, whether it be personal information about clients or banking information.





NSL-KDD Dataset – User 2 Root (U2R) Attacks

- U2R is an attack that starts off with a normal user account and tries to gain access to the system or network, as a super-user (root).
- The attacker attempts to exploit the vulnerabilities in a system to gain root privileges/access.
- Examples: Perl, Load Module and Eject attacks.



NSL-KDD Dataset – Remote 2 Local (R2L) Attacks

- R2L is an attack where intruders sends a set of packets to another computer or server over a network where they do not have permission to access as a local user.
- Examples of R2L include guessing passwords, ftp writes and IMAP.
- Notice the difference between U2R and R2L. **Can you explain?**

How are the attack categories listed on the dataset?

- In column number 42 in the dataset, the attack is represented by its sub-category (and not the parent category). So, you will find **neptune** in the activity type column instead of (DoS).

Classes:	DoS	Probe	U2R	R2L
Sub-Classes:	• apache2 • back • land • neptune • mailbomb • pod • processtable • smurf • teardrop • udpstorm • worm	• ipsweep • mscan • nmap • portsweep • saint • satan	• buffer_overflow • loadmodule • perl • ps • rootkit • sqlattack • xterm	• ftp_write • guess_passwd • httptunnel • imap • multihop • named • phf • sendmail • Snmpgetattack • spy • snmpguess • warezclient • warezmaster • xlock • xsnoop
Total:	11	6	7	15



NSL-KDD Dataset – 41 Input Features

The 41 features in every traffic input can be broken down into four categories:

- Intrinsic features
- Content-based Features
- Time-based Features
- Host-based Features

NSL-KDD Dataset – 41 Input Features - Intrinsic

Intrinsic features can be derived from the header of the packet without looking into the payload. They hold the basic information about the packet. This category contains features 1–9.

#	Feature Name	Description	Type	Value Type	Ranges (Between both train and test)
1	Duration	Length of time duration of the connection	Continuous	Integers	0 - 54451
2	Protocol Type	Protocol used in the connection	Categorical	Strings	
3	Service	Destination network service used	Categorical	Strings	
4	Flag	Status of the connection – Normal or Error	Categorical	Strings	
5	Src Bytes	single connection	Continuous	Integers	0 - 1379963888
6	Dst Bytes	single connection	Continuous	Integers	0 - 309937401
7		If source and destination IP addresses and port numbers are equal then, this variable takes value 1 else 0	Binary	Integers	{ 0 , 1 }
8	Wrong Fragment	Total number of wrong fragments in this connection	Discrete	Integers	{ 0,1,3 }
9	Urgent	are packets with the urgent bit activated	Discrete	Integers	0 - 3

NSL-KDD Dataset – 41 Input Features - Content

Content features hold information about the original packets, as they are sent in multiple pieces rather than one. With this information, the system can access the payload. This category contains features 10–22.

#	Feature Name	Description	Type	Value Type	Ranges (Between both train and test)
10	Hot	system directory, creating programs and executing programs	Continuous	Integers	0 - 101
11	Logins	Count of failed login attempts	Continuous	Integers	0 - 4
12	Logged In	Login Status : 1 if successfully logged in; 0 otherwise	Binary	Integers	{ 0 , 1 }
13	Compromised	Number of "compromised" conditions	Continuous	Integers	0 - 7479
14	Root Shell	1 if root shell is obtained; 0 otherwise	Binary	Integers	{ 0 , 1 }
15	Su Attempted	1 if "su root" command attempted or used; 0 otherwise	(Dataset	Integers	0 - 2
16	Num Root	as a root in the connection	Continuous	Integers	0 - 7468
17	Creations	Number of file creation operations in the connection	Continuous	Integers	0 - 100
18	Num Shells	Number of shell prompts	Continuous	Integers	0 - 2
19	Files	Number of operations on access control files	Continuous	Integers	0 - 9
20	Cmds	Number of outbound commands in an ftp session	Continuous	Integers	{ 0 }
21	Is Hot Logins	1 if the login belongs to the "hot" list i.e., root or admin; else 0	Binary	Integers	{ 0 , 1 }
22	Is Guest Login	1 if the login is a "guest" login; 0 otherwise	Binary	Integers	{ 0 , 1 }

NSL-KDD Dataset – 41 Input Features – Time-based

Time-based features hold the analysis of the traffic input over a two-second window and contains information like how many connections it attempted to make to the same host. These features are mostly counts and rates rather than information about the content of the traffic input. This category contains features 23–31.

#	Feature Name	Description	Type	Value Type	Ranges (Between both train and test)
23	Count	current connection in the past two seconds	Discrete	Integers	0 - 511
24	Srv Count	the current connection in the past two seconds	Discrete	Integers	0 - 511
25	Serror Rate	s0, s1, s2 or s3, among the connections aggregated in count	Discrete	(hundredths of	0 - 1
26	Srv Serror Rate	s0, s1, s2 or s3, among the connections aggregated in	Discrete	(hundredths of	0 - 1
27	Rerror Rate	REJ, among the connections aggregated in count (23)	Discrete	(hundredths of	0 - 1
28	Srv Rerror Rate	REJ, among the connections aggregated in srv_count (24)	Discrete	(hundredths of	0 - 1
29	Same Srv Rate	among the connections aggregated in count (23)	Discrete	(hundredths of	0 - 1
30	Diff Srv Rate	among the connections aggregated in count (23)	Discrete	(hundredths of	0 - 1
31	Rate	destination machines among the connections aggregated in	Discrete	(hundredths of	0 - 1

NSL-KDD Dataset – 41 Input Features – Host-based

Host-based features are similar to Time-based features, except instead of analyzing over a 2-second window, it analyzes over a series of connections made (how many requests made to the same host over x-number of connections). These features are designed to access attacks, which span longer than a two-second window time-span. This category contains features 32–41.

#	Feature Name	Description	Type	Value Type	Ranges (Between both train and test)
32	Dst Host Count	address	Discrete	Integers	0 - 255
33	Count	Number of connections having the same port number	Discrete	Integers	0 - 255
34	Srv Rate	among the connections aggregated in dst_host_count (32)	Discrete	(hundredths of	0 - 1
35	Rate	among the connections aggregated in dst_host_count (32)	Discrete	(hundredths of	0 - 1
36	Src Port Rate	port, among the connections aggregated in dst_host_srv_count	Discrete	(hundredths of	0 - 1
37	Host Rate	destination machines, among the connections aggregated in	Discrete	(hundredths of	0 - 1
38	Rate	s0, s1, s2 or s3, among the connections aggregated in	Discrete	(hundredths of	0 - 1
39	Serror Rate	s1, s2 or s3, among the connections aggregated in	Discrete	(hundredths of	0 - 1
40	Rate	REJ, among the connections aggregated in dst_host_count	Discrete	(hundredths of	0 - 1
41	Rerror Rate	REJ, among the connections aggregated in	Discrete	(hundredths of	0 - 1

Waitlisted Students

